PCAP LAB 7

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Questions:

1. Write a program in CUDA to count the number of times a given word is repeated in a sentence. (Use Atomic function)

2. Write a CUDA program that reads a string ***S*** and produces the string ***RS*** as follows:
   Input string ***S***: PCAP Output string ***RS***: PCAP**PCA**PC**P**
   Note: Each work item copies required number of characters from *S* in *RS*.

Answers:

A

CUDA Program — Count Occurrences of a Word in a String

Program

```
#include "cuda_runtime.h"
#include "device_launch_parameters.h"

#include <stdio.h>
#include <string.h>
#include <stdlib.h>

__global__ void countWords(char *sent, char *word, int n, int m, int *count)
{
    int i = threadIdx.x + blockIdx.x * blockDim.x;

    if (i <= n - m)
    {
```

```
            int match = 1;

            for (int j = 0; j < m; j++)
            {
                if (sent[i + j] != word[j])
                {
                    match = 0;
                    break;
                }
            }

            if (match)
            {
                atomicAdd(count, 1);
            }
        }
    }

int main(void)
{
    char *d_sent, *d_word;
    int *d_count;

    char sent[200];
    char word[100];
    int count = 0;

    printf("Enter sentence and word: ");
    scanf("%s %s", sent, word);

    cudaMalloc((void **)Cd_sent, 200 * sizeof(char));
    cudaMalloc((void **)Cd_word, 100 * sizeof(char));
    cudaMalloc((void **)Cd_count, sizeof(int));

    cudaMemcpy(d_sent, sent, 200 * sizeof(char), cudaMemcpyHostToDevice);
    cudaMemcpy(d_word, word, 100 * sizeof(char), cudaMemcpyHostToDevice);
    cudaMemcpy(d_count, Ccount, sizeof(int), cudaMemcpyHostToDevice);
```

```
    int n = strlen(sent);
    int m = strlen(word);

    int threads = 256;
    int blocks = (n + threads - 1) / threads;

    countWords<<<blocks, threads>>>(d_sent, d_word, n, m, d_count);

    cudaMemcpy(Ccount, d_count, sizeof(int), cudaMemcpyDeviceToHost);

    printf("Count of word: %d\n", count);

    cudaFree(d_sent);
    cudaFree(d_word);
    cudaFree(d_count);

    return 0;
}
```

Sample Run 1

Input
Enter sentence and word: banana ana

Output
Count of word: 2

Sample Run 2

Input
Enter sentence and word: hellohello hello

Output
Count of word: 2

CUDA Program – Produce strings

Example:

S = PCAP
RS = PCAP + PCA + PC + P

Final result:

PCAPP CAPCP

Program

```
#include "cuda_runtime.h"
#include "device_launch_parameters.h"

#include <string.h>
#include <stdio.h>
#include <stdlib.h>

__global__ void prodStrings(char *string, char *out, int n)
{
  int i = threadIdx.x + blockIdx.x * blockDim.x;

  if (i < n)
  {
    int start = (i * (2 * n - i + 1)) / 2;

    for (int j = start; j < start + n - i; j++)
    {
      out[j] = string[j - start];
    }
  }
}

int main(void)
{
  char word[100];
```

```
    printf("Enter word: ");
    scanf("%s", word);

    int n = strlen(word);
    int resLen = (n * (n + 1)) / 2;

    char res[resLen + 1];

    char *d_word, *d_res;

    cudaMalloc((void **)Cd_word, n * sizeof(char));
    cudaMalloc((void **)Cd_res, resLen * sizeof(char));

    cudaMemcpy(d_word, word, n * sizeof(char), cudaMemcpyHostToDevice);

    prodStrings<<<1, n>>>(d_word, d_res, n);

    cudaMemcpy(res, d_res, resLen * sizeof(char), cudaMemcpyDeviceToHost);

    res[resLen] = '\0';

    printf("Result: %s\n", res);

    cudaFree(d_res);
    cudaFree(d_word);

    return 0;
}
```

Sample Run 1

Input
Enter word: PCAP

Output
Result: PCAPPCAPCP

Actual string stored:

PCAPPCA PCP

Sample Run 2

Input
```
Enter word: ABC
```

Output
Result: ABCABA

Actual string:

ABCABA

Explanation:

ABC + AB + A = ABCABA